

SRAI

Professional Obligations

*Canonical Lesson for Official Statisticians Working with Artificial Intelligence***OS-A03-A01 | Controlled candidate v0.1**

Official Statistics and AI Pathway

27 September 2026

Lesson control	Approved basis
Module	OS-A03 Professional Obligations
Learning outcome	Apply impartiality, confidentiality, transparency, quality and accountability to scenarios.
Required product	Professional judgement memorandum
AI-enabled performance	Use AI to analyze professional scenarios and draft judgement records quickly, while independently checking obligations, risks and accountable decisions.
Audience	Official statisticians, methodologists, data stewards, managers and enabling professionals

How to Use This Lesson

This lesson is an advanced professional reference and a practical decision guide. It does not offer a universal ethics checklist or replace national law. It teaches a disciplined way to locate authority, establish facts, recognize competing obligations, protect information, verify AI-assisted reasoning and record an accountable statistical decision. The core conclusion is simple: AI may accelerate analysis and drafting, but it cannot inherit the official statistician's mandate, professional independence or answerability for a statistical product.

Learning outcomes

1. Identify applicable professional obligations, accountable authorities, affected users and consequences in an AI-assisted statistical scenario.
2. Separate verified facts, governing requirements, professional interpretations, assumptions and unresolved uncertainty.
3. Determine whether information may be used in a proposed AI environment and document exclusions or controls.
4. Compare conventional and AI-assisted scenario analysis using the same acceptance standard and quality-adjusted time.
5. Challenge AI-generated reasoning for omission, false authority, leakage, bias and overconfidence.
6. Issue and defend a professional judgement memorandum that accepts, conditions, restricts, rejects or escalates the proposed action.

The recurring decision sequence

Step	Question	Evidence
1 Authority	Who may decide, advise, review and release?	Mandate, delegation, policy and named owner
2 Facts	What is verified, inferred, assumed or disputed?	Claim ledger and source hierarchy
3 Information	What may enter which environment?	Classification, minimization and retention decision
4 Obligations	Which duties apply and where do they conflict?	Obligation map and affected-user analysis
5 Options	What can be accepted, conditioned, restricted, rejected or escalated?	Options and consequence analysis
6 Verification	What independent checks challenge the AI-assisted analysis?	Tests, corrections and residual uncertainty
7 Decision	What must happen, by whom, under which stopping rule?	Professional judgement memorandum

Notation and source conventions

Bracketed references such as [1] point to the consolidated reference list. United Nations principles and national law are normative authorities within their scope. Quality frameworks describe institutional arrangements and evidence. Professional-ethics declarations guide judgement but do not displace binding law. Explanatory literature supports analysis and must not be presented as legal authority.

Chapter 1 Why Professional Obligations Govern AI Assisted Statistics

Official statistics are public infrastructure. Their value depends not only on technical accuracy but also on legitimate authority, impartial production, protected information, transparent methods and accountable release. Artificial intelligence changes the speed, scale and opacity of statistical work, but it does not change these conditions of legitimacy.

Trust is an institutional achievement

Trust should not be treated as public approval of every estimate. A trustworthy statistical system is one in which users have justified reasons to believe that outputs are produced under stable professional rules, appropriate methods and visible correction mechanisms. The United Nations Fundamental Principles place relevance, impartiality, professional standards, transparency and confidentiality within one system of obligations [1]. Each obligation supports the credibility of the others.

AI can improve timeliness, consistency and access to complex information. It can also create a polished output before the underlying authority, data quality or evidentiary basis has been established. This changes the order of risk. A team may now encounter a plausible answer earlier than it encounters the reasons for believing or rejecting it. Professional obligation therefore operates before, during and after tool use: before use to define authority and safe inputs, during use to verify outputs, and after use to decide and document responsibility.

Performance is necessary but insufficient

A classifier may achieve a high average score and still violate the intended classification, fail a subgroup or rely on information that was not lawfully available. A language model may draft an accessible release and still alter an approved interpretation. The professional question is not whether the tool worked in the abstract. It is whether the complete statistical product is fit for its authorized purpose under the applicable quality, confidentiality and accountability conditions.

This lesson uses a non-compensatory logic. Productivity does not cancel a confidentiality failure. Accuracy on common cases does not cancel systematic harm to a small population when that population matters to the use. A deadline does not create release authority. These are not anti-innovation rules. They define the conditions under which innovation can be relied upon in public statistics.

The statistician remains the decision bearer

The official statistician may delegate calculation, search, coding assistance or drafting. The statistician cannot delegate the institutional mandate or the duty to explain why evidence was accepted. Human oversight is meaningful only when the reviewer can inspect relevant inputs and evidence, challenge the method, require correction and stop release. A nominal human signature after an opaque process is not accountability.

The practical product of this module is therefore a professional judgement memorandum. The memorandum does not merely describe a concern. It establishes authority, facts, obligations, options, verification, decision, safeguards and review. It converts professional judgement from an intuition into a record that another qualified person can scrutinize.

Technical success	Professional question	Required record
Fast draft	Did it preserve approved facts, meaning and uncertainty?	Source comparison and correction log
High accuracy	Is performance fit for the intended use and relevant groups?	Benchmark and subgroup evidence
Automated workflow	Can an authorized reviewer inspect, challenge and override it?	Responsibility and review map
Useful output	Was protected information handled in an approved environment?	Data-handling decision

Worked professional situation

Kintaba's labour-statistics team receives an AI-generated briefing that is factually fluent and delivered in minutes. The draft contains the correct headline rate but describes a non-significant quarterly change as a clear deterioration and omits a break in the sampling frame. The professional failure is not simply a wording error. It concerns quality, transparency and accountability because the draft could change public interpretation. The correct response is to stop publication, restore the approved significance and revision language, document the AI failure and determine whether the instruction or review procedure must change.

Verification and decision checks

- State the authorized statistical purpose before evaluating the tool.
- Apply the same acceptance criteria to conventional and AI-assisted outputs.
- Treat plausibility as a reason to inspect, not as evidence of correctness.
- Name the person who can require correction and stop release.

Study questions

7. Why can a statistically accurate number still support an unacceptable public product?
8. Which professional obligations are non-compensatory in your institution?
9. What evidence would make human oversight meaningful rather than ceremonial?

Chapter sources [1], [2], [4], [8]

Chapter 2 The Accountable Official Statistician

Professional accountability begins with role clarity. AI systems can perform tasks, but they do not occupy statutory office, hold a delegation, experience professional sanction or answer to users. The accountable statistician must know which decisions belong to the institution, which belong to a designated official and which may be supported by a tool.

Mandate role and task are different

A mandate authorizes an institution to collect, process or release statistics. A role allocates responsibility within that mandate. A task is an activity that may be performed by a person or system. Confusion arises when the performance of a task is treated as possession of the role. An AI system may flag unusual values; it does not acquire authority to revise them. It may propose a classification; it does not determine the official concept.

The responsibility map should distinguish the statistical owner, data owner, methodologist, processor, confidentiality officer, system owner, approving official and release authority. Small institutions may combine roles, but they should still identify the distinct functions and seek independent challenge where the same person would otherwise develop and approve a consequential output.

Accountability requires effective control

A person is not meaningfully accountable for an output they cannot inspect, understand sufficiently, challenge or change. Effective control depends on access to evidence, adequate competence, time for review and practical power to stop the process. If a vendor withholds logs or if an internal system offers only a confidence score without the relevant cases, the institution may lack the conditions needed to assign professional approval.

The appropriate response is not always to reject AI. The workflow may be redesigned so the system makes recommendations on bounded cases, sends uncertain or consequential cases to human review and retains enough evidence for audit. Accountability becomes an architectural requirement: it shapes interfaces, logs, sampling, override rights, incident procedures and procurement terms.

Advice approval and release

AI-assisted work often passes through several decisions. A tool may advise; an analyst may edit; a methodologist may approve the method; a manager may authorize operational use; and a chief statistician may hold release authority. These decisions should not be collapsed into one vague statement that 'a human reviewed it.' The record should state what each reviewer examined and what decision each was empowered to make.

Where the evidence is inadequate, the professional response is to qualify, restrict or escalate. Escalation is not a failure of competence. It is evidence that the analyst recognizes the boundary of their authority or knowledge. Conversely, signing a result to avoid delay when essential evidence is unavailable is an accountability failure even if the result later proves correct.

Function	Accountable question	AI may support	AI must not decide
Method	Does the approach represent the statistical concept?	Explain code and compare alternatives	Official concept or tolerance without approval
Data	May these inputs be used here?	Inventory fields and identify apparent sensitivity	Legal basis or confidentiality classification
Quality	Does evidence meet the use threshold?	Calculate metrics and surface failures	Acceptance of residual risk
Release	Should the product be disseminated now?	Check consistency and draft text	Release authority or embargo exception

Worked professional situation

A regional office uses an AI coding service for occupation descriptions. The service returns codes and confidence scores, but the central methodologist cannot see the training examples or subgroup performance. The office manager asks the methodologist to approve because a human will sign the monthly table. The signature would be ceremonial: the reviewer lacks evidence needed to challenge the coding. A defensible decision conditions approval on a benchmark sample, error analysis by language and occupation group, retained case-level evidence and an escalation threshold. Until then, the output may support exploration but not official production.

Verification and decision checks

- Identify the statutory or delegated owner of the statistical product.
- Separate task execution, method approval, operational authorization and release.
- Confirm that reviewers possess evidence, competence, time and stopping power.
- Record restrictions when evidence supports assistance but not consequential use.

Study questions

10. What decisions in your workflow are routinely hidden inside the phrase human in the loop?
11. When can combined roles be acceptable in a small institution?
12. Which evidence rights should be included in an AI supplier agreement?

Chapter sources [1], [3], [5], [8]

Chapter 3 Normative Architecture of Official Statistics

Professional judgement is disciplined by a hierarchy of authority. The official statistician must distinguish binding law, institutional policy, international principles, quality frameworks, professional ethics and explanatory scholarship. AI systems frequently flatten these sources into equally confident text; the professional must restore their different status.

The Fundamental Principles as a system

The United Nations Fundamental Principles of Official Statistics connect democratic need, impartiality, scientific standards, source accountability, confidentiality, coordination and international cooperation [1]. They should be read together. Principle 1's commitment to relevance, impartiality and equal access is supported by professional methods, transparent presentation, protected individual data and institutional coordination. Selective invocation can create false solutions, such as claiming transparency requires disclosure of confidential records.

The principles establish direction but often require national law and policy to determine concrete authority. A national statistics act may define the chief statistician's powers, secrecy offences, pre-release arrangements or mandatory data access. A release policy may define embargoes. A security policy may define approved computing environments. The memorandum must cite the controlling local authority when a decision depends on it.

Quality and ethics add different functions

The UN National Quality Assurance Framework treats quality as an institutional system rather than a final accuracy check [2]. It connects user needs, processes, people, methods, outputs and continuous improvement. The ISI Declaration on Professional Ethics emphasizes shared values and informed individual judgement [4]. Quality frameworks ask whether processes reliably produce fit-for-use outputs; ethics frameworks help professionals reason about responsibilities and conflicts. Neither should be misrepresented as national law.

Codes of practice, including regional frameworks, can specify professional independence, coordination, resources, methodology, confidentiality and accessibility in operational terms [3][5]. They are useful comparative evidence when designing institutional controls. Their formal authority depends on the jurisdiction and organization.

A source hierarchy prevents fabricated authority

Generative AI may invent section numbers, merge jurisdictions or attribute a principle to the wrong body. A reliable workflow begins with a controlled source register containing the title, issuing body, version, jurisdiction, status, URL or repository location and relevant provisions. Retrieval should be restricted to approved sources where possible. Every material claim in a judgement record should point to a source or be marked as professional interpretation.

When sources conflict, the analyst should not ask AI to choose which law prevails. The conflict must be routed to an authorized legal or policy function. AI may summarize the competing texts and identify factual differences, but precedence and legal interpretation remain outside the model's authority.

Source type	Typical authority	Proper use	Warning
National law	Binding in jurisdiction	Mandate, secrecy, access, offences	Requires authorized interpretation
Institutional policy	Binding internally when valid	Release, security, retention, review	Confirm version and scope
UN principles	Normative global standard	Professional architecture and trust	Not a substitute for local law
Quality framework	Methodological and organizational guidance	Processes, evidence and assurance	Adapt to context
Ethics declaration	Professional reasoning framework	Values, conflicts and conduct	Do not cite as statute

Worked professional situation

An AI assistant tells a statistician that 'UN Principle 6 requires open publication of all underlying microdata.' The claim is false: Principle 6 concerns correction of misuse, while confidentiality is protected under Principle 6 of some other frameworks and Principle 6 is not a universal microdata-access rule. The professional checks the official UN text, separates open-data policy from statistical confidentiality and records that only disclosure-controlled access authorized by national arrangements may be offered. The incident is added to the prompt test set because it demonstrates fabricated authority.

Verification and decision checks

- Record issuing body, version, jurisdiction and normative status.
- Verify every quoted provision in the authoritative source.
- Separate legal conclusion from statistical or ethical interpretation.
- Escalate unresolved precedence rather than allowing the tool to resolve it.

Study questions

13. How should an analyst proceed when a regional code and national policy use different terminology?
14. What fields belong in a controlled authority register?
15. Why is a correct ethical conclusion still defective if supported by a fabricated citation?

Chapter sources [1], [2], [3], [4], [5]

Chapter 4 Ethical Reasoning and Compliance

Compliance asks whether a rule has been followed. Professional reasoning asks what an accountable statistician should do when rules are incomplete, competing or open to interpretation. Official-statistics work requires both: rules establish boundaries, while principled judgement addresses circumstances that rules did not anticipate.

Rules principles and discretion

A rule can prohibit entering identifiable records into an unapproved service. A principle explains why: statistical confidentiality, purpose limitation and public trust. Discretion is the authorized space in which the professional decides how to apply these requirements to a new situation. Discretion is not personal preference. It must be exercised through relevant evidence, consistent reasoning, proportionality and a record that another qualified person can review.

AI can assist by organizing facts, generating alternatives and identifying apparent tensions. It tends, however, to complete missing context with general assumptions. A good task specification requires it to label uncertainty, distinguish provided authority from general background and refuse to invent a rule. The reviewer then checks whether the analysis omitted affected groups, institutional dependencies or practical implementation constraints.

The danger of checklist ethics

Checklists are valuable for memory and minimum controls. They become dangerous when a checked box is treated as proof that the underlying duty has been satisfied. A confidentiality box may be checked even though the analyst misunderstood data retention. A fairness box may be checked after computing an irrelevant metric. The professional memorandum must therefore connect each conclusion to evidence, consequence and authority.

A robust analysis also considers options. Binary approve-or-reject framing can conceal a safer restricted use, a redesigned environment, a provisional release or a requirement for additional evidence. Conversely, adding conditions should not be used to rescue an option that fails a mandatory gate.

Reasonableness and consistency

Professional judgement should be consistent across similar cases unless a material difference justifies departure. Institutions can improve consistency through scenario libraries, calibration exercises and peer review. Consistency does not mean uniform outcomes when context differs; it means that relevant differences are identified and irrelevant differences do not drive the result.

The memorandum makes consistency testable. It states the facts relied upon, the obligations triggered, the options considered and the residual risk accepted. Future reviewers can compare cases, detect drift and update policy when recurring ambiguity reveals a systemic gap.

Weak reasoning	Why it fails	Stronger reasoning
The tool is ethical	Attributes moral agency and hides context	The proposed use meets specified authority, data, quality and oversight conditions
A human reviews it	Does not describe effective control	Named reviewer inspects defined evidence and can override or stop
No law prohibits it	Confuses absence of prohibition with professional acceptability	Mandate, purpose, necessity, safeguards and consequences are established
Benefits outweigh risks	Too aggregated; may compensate a failed gate	Mandatory gates pass, then residual benefits and risks are compared

Worked professional situation

A team proposes using an AI assistant to classify public consultation responses. No policy explicitly mentions this tool. A checklist-only review marks the data as public and approves. A professional review asks whether respondents expected automated processing, whether sensitive information appears in free text, whether classification affects representation of minority views, whether the model can be audited and whether the method will be disclosed. The decision may permit a pilot on redacted text with double coding and disagreement review while prohibiting automated deletion or final thematic weighting.

Verification and decision checks

- Identify mandatory gates before balancing residual considerations.
- Connect every checked control to evidence and an accountable owner.
- Consider restricted and redesigned options, not only approve or reject.
- Compare the reasoning with analogous prior decisions and explain departures.

Study questions

16. When does discretion become unauthorized policy making?
17. How can scenario calibration improve consistency without suppressing context?
18. Which conditions cannot rescue a failed confidentiality gate?

Chapter sources [1], [4], [8]

Chapter 5 Impartiality in Statistical Purpose and Method

Impartiality requires official statistics to be developed, produced and disseminated on an objective basis determined by professional considerations. In AI-assisted work, impartiality applies to problem framing, data selection, target definition, model choice, evaluation, narrative and access, not merely to the final wording.

Bias begins before the model

A system can be computationally neutral with respect to its instruction and still implement a partial institutional question. If a use case is framed only around reducing unfavorable estimates, excluding difficult populations or accelerating a preferred release, technical optimization may reinforce improper purpose. The first impartiality control is therefore a clear statistical need, authorized user purpose and stated consequences of error.

Training data and retrieval sources also structure what the AI can see. Historical labels may reflect enforcement intensity, administrative priorities or classification practices rather than the concept of interest. Retrieval systems may overrepresent well-documented institutions or dominant languages. Impartiality requires analysis of the data-generating process and not only mathematical parity in outputs.

Independence in method and interpretation

Professional independence protects decisions about concepts, sources, methods, timing and interpretation from inappropriate influence. Management may set legitimate priorities and resource constraints; it should not direct a statistical conclusion. AI introduces a subtler influence: a default model, vendor interface or prompt template may silently determine categories, thresholds or narrative emphasis. The official statistician must be able to identify and change those defaults.

Narrative generation deserves particular scrutiny. Models tend to produce coherent causal language and may amplify the framing in the prompt. A neutral review asks whether alternative interpretations were tested, whether uncertainty is symmetrical and whether favorable and unfavorable evidence receive comparable treatment.

Equal access and differentiated need

Impartiality supports equal access to official statistical information, but equal access does not require identical communication for every user. Accessible language, translated material and formats for persons with disabilities may be necessary to achieve meaningful access. The distinction is between justified adaptation and privileged substantive access.

Pre-release arrangements, embargoes and stakeholder consultation must follow authorized policy. AI-assisted personalization should not allow a favored user to receive unreleased statistics or a different evidentiary standard. Logs and publication workflows should make exceptional access visible.

Decision point	Impartiality risk	Evidence needed
Use-case framing	Problem chosen to produce a desired result	User need, public value and rejected alternatives
Data selection	Convenient sources exclude relevant groups	Coverage and missingness profile
Threshold	Default optimizes average performance only	Loss function, subgroup consequences and tolerance approval
Narrative	Prompt embeds preferred interpretation	Alternative prompts and source-grounded review
Access	Personalization creates privileged release	Release policy and access logs

Worked professional situation

A ministry asks the NSO to use an AI narrative tool to emphasize improvements in food prices before an election. The underlying index is correct, but the requested framing selects a short favorable interval and ignores longer-term volatility. The statistician separates the legitimate need for accessible explanation from the inappropriate instruction to promote a conclusion. The final release uses the approved comparison periods, explains seasonal and revision factors, and records the attempted interference through the institutional escalation route.

Verification and decision checks

- Test whether the statistical purpose is legitimate before optimizing the workflow.
- Inspect source coverage, labels, language and documentation gaps.
- Challenge defaults that embed thresholds or categories without professional approval.
- Apply release and access rules consistently and retain exception logs.

Study questions

19. Can a model be unbiased while the statistical use remains partial? Explain.
20. When is differentiated communication compatible with equal access?
21. What evidence distinguishes management priority setting from improper methodological interference?

Chapter sources [1], [3], [5], [8]

Chapter 6 Professional Independence and Interference

Professional independence is the capacity of statistical authorities to make and communicate methodological and release decisions without improper political, commercial or administrative direction. It is not isolation from users or government. It is a governance condition that protects the credibility of statistics used by all parties.

Forms of interference

Interference can be explicit: an instruction to change an estimate, suppress a table or delay a release. It can also operate through budget pressure, access restrictions, personnel decisions, selective briefings, repeated demands for favorable re-analysis or tool procurement that removes methodological control. AI can make interference less visible when the desired result is embedded in a prompt, ranking objective or proprietary default.

A professional should document the request, the statistical issue, the applicable policy, the response and the escalation. Documentation must be accurate and proportionate; it should not itself disclose confidential deliberation beyond authorized channels. The purpose is to preserve an evidence trail and enable institutional protection, not to personalize disagreement.

Urgency and emergency production

Emergency statistics may require accelerated methods, provisional classifications or reduced detail. Urgency can justify a different approved process; it does not erase professional standards. The product should state its provisional status, known limitations, revision plan and appropriate use. A time-limited derogation should have an owner and expiry.

AI may help produce faster summaries, nowcasts or issue triage. The statistician should identify which controls can be performed before release, which must follow immediately and which failures require stopping despite urgency. A transparent provisional product can be professionally stronger than an apparently final product supported by unverified automation.

Institutional safeguards

Independence is strengthened by published release calendars, transparent revision policies, clear appointment and dismissal arrangements, methodological committees, documented pre-release access and the authority to comment on misuse. Technical architecture can also help: version control, immutable logs, reproducible pipelines and separation of development from approval make inappropriate changes easier to detect.

Individuals should not be expected to carry systemic risk alone. Escalation routes, peer bodies and professional associations can support difficult cases. Training should rehearse interference scenarios so that staff know whom to contact and what evidence to retain before a crisis.

Pressure	Legitimate response	Boundary
Earlier release	Use an approved emergency procedure with provisional status	Do not conceal incomplete validation

Pressure	Legitimate response	Boundary
Alternative analysis	Perform if methodologically justified and label it clearly	Do not replace the official method to obtain a desired result
Communication request	Improve accessibility and context	Do not distort uncertainty or comparison periods
Budget constraint	Document risk and prioritize critical controls	Do not certify checks that were not performed

Worked professional situation

A ministerial office requests a two-day delay in an unemployment release, arguing that the minister needs time to prepare a response. The release calendar and pre-release policy do not authorize the delay. An AI assistant drafts a conciliatory reply but incorrectly suggests that reputational sensitivity is a valid statistical exemption. The statistician verifies the policy, removes the invented exemption, informs the designated release authority and retains the request and decision in the controlled record. The release proceeds on schedule.

Verification and decision checks

- Distinguish legitimate priority or emergency decisions from instructions about statistical conclusions.
- Verify any claimed exemption in the controlling release policy.
- Record material interference through the authorized route.
- Use reproducibility and change logs to make unauthorized alteration detectable.

Study questions

22. How can procurement decisions weaken professional independence?
23. What controls may be deferred in an emergency and who may authorize that decision?
24. Why should interference be treated as an institutional risk rather than an individual moral test?

Chapter sources [1], [3], [5], [13]

Chapter 7 Confidentiality and Statistical Secrecy

Statistical confidentiality is not a courtesy owed only to named respondents. It is a legal and professional obligation governing information collected or obtained for statistical purposes, including direct identifiers, attributes, combinations, business information and outputs from which a unit may be inferred.

Direct and inferential disclosure

Direct disclosure occurs when a person or organization is explicitly identified. Inferential disclosure occurs when attributes can be learned with high confidence from combinations of data, external information or model outputs. AI systems increase linkage and inference capacity, so removal of names is rarely sufficient evidence of safety. Free text, rare categories, geography, timestamps and embeddings can retain identifying information.

Confidentiality decisions should consider the data, the environment, the user, the purpose and the output together. The same file may be safe in a controlled research environment and unsafe in a public generative service. The classification should be made before data entry, not after the tool has produced an output.

Purpose and access limitation

Access to data for one statistical purpose does not automatically authorize use for tool evaluation, model training or vendor support. The institution should establish legal basis, approved purpose, minimum necessary fields, user authorization, location, retention, onward transfer and deletion. A model provider's general assurance is not a substitute for the institution's decision.

Synthetic or public data should be used for basic training and prompt development. When realistic structure is required, carefully designed synthetic cases can preserve error mechanisms without representing actual persons. If confidential data are necessary, the activity requires an approved secure environment, documented access and appropriate output control.

Confidentiality and transparency

Transparency about methods does not require revealing confidential information. An institution can explain categories, validation procedures, model limitations and disclosure controls while withholding protected details. Conversely, confidentiality should not be used to conceal methodological weaknesses that can be described safely.

The professional memorandum should state what information was excluded from AI interaction and why. This makes absence a deliberate control rather than a hidden limitation. It should also state what evidence remains available to authorized reviewers.

Layer	Example risk	Control
Input	Names, rare combinations, free text, coordinates	Minimize, transform, synthesize or use secure environment

Layer	Example risk	Control
Prompt	Sensitive context embedded in instructions	Use approved templates and prohibited-field checks
Service	Retention, training use, foreign processing, support access	Tool clearance and contractual controls
Output	Memorized text or small-cell inference	Output checking and disclosure control
Evidence	Logs and screenshots reproduce protected data	Restricted retention and access

Worked professional situation

A health-statistics analyst wants AI help grouping free-text causes recorded in a confidential mortality file. Names were removed, but the text includes villages, occupations and unusual events that may identify individuals. The analyst does not upload the file to a public tool. Instead, the team develops the instruction on synthetic examples, runs the approved model inside the controlled environment, samples difficult cases for expert review and applies output disclosure controls before any aggregate table is released.

Verification and decision checks

- Classify information before it enters any AI workflow.
- Confirm purpose, authority, environment, retention and onward transfer.
- Treat free text, embeddings and outputs as potentially disclosive.
- Retain only the minimum evidence necessary under authorized controls.

Study questions

25. Why is de-identification not a binary property?
26. When can synthetic data support meaningful training without overstating validity?
27. How can an institution be transparent about a confidential workflow?

Chapter sources [1], [2], [3], [5], [9]

Chapter 8 Confidentiality Risks in AI Environments

AI environments introduce data flows that are easy to overlook: prompts may be logged, retrieval documents indexed, outputs retained, telemetry collected and vendor personnel granted support access. Professional confidentiality review must follow the complete system rather than the visible chat window.

Map the information path

A system map should identify the user device, identity service, application, model endpoint, retrieval store, logs, monitoring, backups, support channels and downstream exports. For each component, the institution should know what information is processed, where, for how long, by whom and under what terms. Ambiguity is itself a risk requiring resolution before protected use.

Retrieval-augmented generation can reduce unsupported answers by grounding a model in institutional documents, but it creates an indexed copy and new access paths. Permissions in the retrieval layer must reflect source permissions. A user should not retrieve a confidential document merely because it was embedded in a common index.

Training use and model memory

Institutions should distinguish transient inference from provider training, tuning, human review and service improvement. Contract terms and technical configuration must align. Even when provider training is disabled, prompts may be retained for abuse monitoring or troubleshooting. The retention period and access should be known and accepted.

Model outputs can reproduce sensitive strings from context or expose inferred attributes. Testing should include extraction attempts, prompt injection, indirect disclosure and cross-user isolation where relevant. High-consequence deployment requires evidence from the actual configuration, not generic claims about the model family.

Proportionate tool clearance

Clearance should be use-specific. A service may be approved for public metadata drafting and prohibited for microdata. A local model may reduce external transfer while introducing patching, access-control and model-governance responsibilities. The decision compares complete risks and operational capacity rather than assuming cloud or local deployment is inherently safe.

Users need clear labels and friction at the point of use. Training alone is insufficient when copying protected text is technically easy. Data-loss prevention, restricted connectors, access controls and approved templates can make the safe action the normal action.

Question	Evidence	Failure response
Are prompts retained?	Configuration and contractual retention term	Exclude protected data until resolved
Can inputs train the service?	Provider term and tenant setting	Disable or reject use
Where is retrieval content stored?	Architecture and data-location record	Restrict index and align permissions

Question	Evidence	Failure response
Who can inspect logs?	Role and support-access record	Apply least privilege and monitoring
Can data be exported or deleted?	Portability and deletion test	Require remediation or alternative

Worked professional situation

Kintaba's NSO considers a vendor assistant connected to its document repository. The demo retrieves a restricted methods note for a user who lacks source access because the index inherited project-level rather than document-level permissions. The team suspends the connector, treats the event as a confidentiality incident, reviews access logs and requires permission-aware retrieval tests before reconsideration. The model's good summaries are irrelevant until access control passes.

Verification and decision checks

- Map every component that receives or derives information.
- Verify tenant settings and contractual terms in the actual configuration.
- Test retrieval permissions and prompt-injection boundaries.
- Define incident containment, deletion and evidence-preservation procedures.

Study questions

28. What hidden data flows exist in a typical enterprise assistant?
29. Why can a local model still create serious confidentiality risk?
30. Which tool-clearance evidence should expire when a provider changes its service?

Chapter sources [5], [8], [9], [14]

Chapter 9 Transparency and Explainability

Transparency in official statistics enables users and reviewers to understand what was produced, for what purpose, from which sources, through which methods and with which limitations. It is not a demand to expose every technical detail or confidential record. It is a duty to provide the information necessary for appropriate interpretation and accountability.

Transparency has audiences

Methodologists need enough detail to reproduce or challenge the process. Managers need evidence about controls, performance and residual risk. Data providers need clarity about use and protection. Public users need concepts, sources, quality and revision information. A single generic explanation rarely serves all audiences. Layered documentation can provide accessible summaries with links to deeper methods and governance records.

AI assistance should be disclosed when it is material to interpretation, assurance or accountability. An institution need not label routine spell checking as if it were a statistical method. It should disclose automated classification, generation or inference when these affect content, uncertainty, comparability or the evidence users need to assess the product.

Explainability is purpose dependent

A global feature-importance chart may help a technical reviewer but fail to explain why a specific record was coded. A natural-language rationale generated after a prediction may be plausible without representing the model's mechanism. Explainability evidence should be selected for the decision: debugging, individual review, subgroup assurance, public method description or appeal.

The official statistician should distinguish explanation of the workflow from explanation of a model output. Workflow explanation may include sources, preprocessing, benchmark, review and stopping rules even when the model is complex. If an output cannot be explained sufficiently for a consequential use, the use may need restriction rather than rhetorical simplification.

Correction and revision

Transparency includes the capacity to correct. AI-assisted systems can change through model updates, retrieval revisions or prompt modifications. Version records should allow the institution to determine which configuration produced an output. Material corrections should be communicated under the revision policy, with an explanation proportionate to user impact.

Over-disclosure can create confusion or security risk. Publishing raw prompts without context may expose controls or protected content and can falsely suggest that a prompt alone determines the result. The transparent object is the controlled procedure: purpose, inputs, model or service class, key instructions, verification, human roles, limitations and change control.

Audience	Needs to know	Normally retained or published
Public user	Meaning, source, quality, revision and material automation	Metadata, methods summary and limitations

Audience	Needs to know	Normally retained or published
Reviewer	Inputs, configuration, tests, errors and corrections	Validation and reproducibility record
Decision body	Benefits, risks, gates, residual uncertainty and ownership	Assurance case and decision minute
Affected person or provider	Relevant processing, safeguards and contestability route	Notice or institutional policy as applicable

Worked professional situation

An AI tool drafts a poverty bulletin from approved tables. The numbers are correct, but the draft attributes the change to employment growth, a causal claim not established by the tables. The analyst removes the claim, adds the approved uncertainty and revision language, and records the correction. The public note describes AI as a drafting aid under analyst review because the generation did not determine estimates, while the internal record retains the prompt, original output and verification.

Verification and decision checks

- Identify what each audience needs for appropriate interpretation.
- Do not present generated rationales as verified model explanations.
- Disclose material automation and retain deeper internal evidence.
- Link corrections to versions, users and the approved revision policy.

Study questions

31. When is disclosure of AI assistance material?
32. How does layered transparency protect both comprehension and confidentiality?
33. What evidence should accompany an opaque model used only for low-consequence triage?

Chapter sources [1], [2], [3], [5], [10]

Chapter 10 Quality as a Professional Obligation

Quality in official statistics is fitness for intended use, not a single accuracy number. Relevance, accuracy, reliability, timeliness, punctuality, coherence, comparability, accessibility and interpretability may all matter. Professional obligation requires the statistician to make these requirements explicit before judging an AI-assisted method.

From dimensions to acceptance rules

A quality dimension becomes operational only when tied to a use, indicator and tolerance. For automated coding, overall accuracy may be supplemented by class-specific error, rare-category performance, stability and reviewer workload. For drafting, factual consistency, preservation of uncertainty, source traceability and accessibility may matter more than textual similarity. The acceptance matrix should state which failures stop use and which can be corrected.

The benchmark must represent the intended use. A convenient historical sample may omit new classifications, changed forms or minority languages. Development data should be separated from locked evaluation evidence. Where no perfect truth exists, independent review, adjudication samples and sensitivity analysis can still support a defensible conclusion.

Quality adjusted time

Efficiency is evaluated after quality. Let baseline effort be T_b and AI-assisted effort include generation T_g , verification T_v and rework T_r . Net time saved is $T_b - (T_g + T_v + T_r)$, but it receives credit only when both outputs meet the same acceptance standard. A fast failed output has no efficiency value and may create additional institutional cost.

Time measurement should capture active work and material waiting separately. The comparison should avoid training effects that unfairly burden one route and should record the level of staff involved. The objective is not to prove AI always saves time; it is to identify tasks and conditions where a controlled method produces reliable gains.

Quality responsibility across the workflow

Quality is created through design, source management, processing, analysis, review and communication. AI can shift error rather than remove it. Automated coding may reduce manual variation while increasing systematic misclassification. Faster drafting may transfer effort to verification. The professional analysis should examine the complete workflow and the users who absorb residual error.

A quality failure should trigger disposition: correct the case, refine the procedure, restrict the use, redesign the benchmark or stop. Silent manual repair can make a weak system appear successful and prevents institutional learning. Corrections must therefore be recorded and analyzed.

Metric	Purpose	Common misuse
Average accuracy	Overall classification agreement	Hides rare or consequential classes
F1 score	Balances precision and recall for a class	Depends on threshold and prevalence

Metric	Purpose	Common misuse
Calibration	Relates predicted confidence to observed outcomes	Does not establish conceptual validity
Stability	Tests sensitivity to rerun or prompt variation	Stable error remains error
Time saved	Measures workflow efficiency	Invalid if quality or review is weakened

Worked professional situation

A new AI coding workflow reduces average processing time by 40 percent and meets the overall accuracy threshold. Review shows that it confuses two agricultural worker categories concentrated in one region, materially affecting an employment indicator. The team rejects the aggregate pass conclusion, restricts automation for those categories, expands the benchmark and recalculates time savings including targeted review. The revised use is slower than the original claim but meets the statistical purpose.

Verification and decision checks

- Define intended use, consequence and quality threshold before testing.
- Use subgroup and failure-specific evidence, not only averages.
- Include verification and rework in efficiency measurement.
- Record corrections so recurrent failure is visible.

Study questions

34. Why can better average accuracy reduce fitness for use?
35. How should verification time be allocated when comparing workflows?
36. What quality dimensions matter for an AI-generated statistical release?

Chapter sources [2], [3], [6], [8]

Chapter 11 Accountability and Human Oversight

Accountability connects authority, evidence, decision and consequence. It requires more than identifying a person after a failure. The workflow must give responsible officials the information and control needed to prevent, detect, correct and explain material errors.

Designing meaningful oversight

Oversight should be matched to consequence and uncertainty. Low-risk drafting from approved facts may use sample review. Automated edits to microdata or classifications affecting estimates may require case-level rules, benchmark monitoring and escalation. High-consequence release or access decisions require explicit approval. The review design should state what is inspected, at what frequency, against which standard and with what stopping authority.

Automation bias can cause reviewers to accept plausible suggestions, while algorithm aversion can cause blanket rejection without evidence. Calibration exercises should expose reviewers to correct and incorrect outputs, including subtle failures. Interfaces should show relevant source evidence and uncertainty rather than only a recommendation and confidence score.

Audit trail and decision trail

A technical log records system events. A decision trail records why people accepted, changed or rejected outputs. Both may be needed. The decision trail should identify material inputs, version, checks, corrections, reviewer and final disposition. It should be proportionate: retaining every transient interaction may create confidentiality and records risks without improving accountability.

Accountability also requires correction routes. Users and staff need a way to report suspected errors. The institution should triage, investigate, correct and communicate material issues. A model update or prompt change should not erase the evidence needed to understand earlier outputs.

Organizational accountability

Responsibility extends beyond the analyst. Leaders authorize resources and risk; procurement secures evidence and exit rights; IT protects environments; data stewards control sources; methodologists define standards; and release authorities approve dissemination. A failure caused by inadequate staffing or unavailable logs should not be framed solely as user error.

The professional judgement memorandum assigns conditions and owners. If a safeguard has no owner, due date or completion evidence, it is an aspiration rather than a control. Residual risk acceptance must be made at the level authorized to bear the consequence.

Oversight mode	Suitable context	Minimum evidence
Full review	Novel or consequential cases	Input, output, sources, checks and decision
Risk-based review	Stable workflow with identifiable high-risk cases	Routing rule, reviewed cases and missed-case analysis

Oversight mode	Suitable context	Minimum evidence
Sample review	Low-consequence high-volume support	Sampling design, error estimate and escalation
Monitoring	Approved production workflow	Quality, subgroup, drift, incident and override indicators

Worked professional situation

An automated editing system proposes changes to reported business turnover. Analysts approve suggestions through a rapid interface. An audit finds that reviewers rarely open the evidence panel because the approve button is prominent and targets are strict. The institution redesigns the interface, requires reasons for material changes, samples approvals, monitors override patterns and adjusts workload. Accountability improves through workflow design rather than another reminder to be careful.

Verification and decision checks

- Match review intensity to consequence, novelty and uncertainty.
- Ensure the reviewer can inspect evidence and override the result.
- Retain a proportionate decision trail as well as technical logs.
- Assign every safeguard an owner, due date and completion test.

Study questions

37. How can an interface weaken human oversight?
38. What is the difference between a technical log and a decision trail?
39. Who should accept residual risk when a safeguard cannot be completed?

Chapter sources [2], [5], [8], [14]

Chapter 12 Fairness Accessibility and Affected Groups

Professional obligations extend to people and institutions affected by statistical processes and products. Fairness in official statistics cannot be reduced to one model metric. It involves representation, measurement, access, burden, error consequences, contestability and the public allocation decisions that statistics inform.

From parity to statistical purpose

Equal error rates may be impossible or inappropriate when groups differ in prevalence, measurement or use. The analyst should begin with the statistical concept and consequences: which errors matter, to whom and under what decision? Metrics then support, rather than replace, judgement. Group definitions must be legitimate, available and protected; fairness analysis can itself create disclosure risk.

Small groups pose a dual challenge. They may be most vulnerable to poor generalization and least able to support precise performance estimates. Uncertainty intervals, qualitative review and targeted data improvement should accompany point estimates. Absence of evidence is not evidence of equal performance.

Accessibility and equal access

Official statistics should be accessible and interpretable. AI can help produce plain language, translation, alternative text and tailored explanations. These outputs require verification because a fluent simplification may change definitions or uncertainty. Accessibility review should involve intended users and should preserve the approved statistical meaning.

Equal access concerns timing and availability as well as format. Personalized services should not create privileged substantive access or exclude people without digital connectivity. Offline and low-bandwidth alternatives may be part of a fair dissemination strategy.

Contestability and correction

When AI supports coding or linkage, affected data providers and internal users may need routes to challenge errors. Official statistics are often aggregate, so individual contestability may differ from administrative decision systems. Still, institutions need mechanisms for source correction, classification disputes and methodological challenge.

The judgement record should identify who is affected, who can detect harm and who can seek correction. A workflow that relies on errors being reported by groups with less access or capacity may reproduce inequality even if the model metric appears acceptable.

Fairness lens	Question	Evidence
Representation	Who is missing or poorly observed?	Coverage, missingness and source-process analysis
Performance	Which groups experience which errors?	Disaggregated metrics with uncertainty
Consequence	How do errors affect use or allocation?	Loss and affected-user analysis

Fairness lens	Question	Evidence
Access	Who can obtain and understand the product?	Channel, language and accessibility testing
Contestability	Who can challenge and correct?	Complaint, correction and review records

Worked professional situation

A multilingual occupation coder performs well overall but has higher rejection and miscoding rates for descriptions submitted in a minority language. The team does not hide the group because sample size is small. It reports uncertainty, conducts expert review, improves language-specific examples and routes affected cases to human coding. Automation is restricted until the subgroup meets the accepted tolerance. The memorandum explains both the evidence and its limits.

Verification and decision checks

- Define affected groups from purpose and consequence, not convenience alone.
- Report uncertainty around subgroup performance.
- Assess accessibility and contestability alongside model metrics.
- Protect sensitive group information during fairness analysis.

Study questions

40. Why may equal error rates be an inadequate fairness objective?
41. How can accessibility adaptation preserve impartiality?
42. What should be done when subgroup evidence is weak but consequences are serious?

Chapter sources [1], [2], [3], [6], [10]

Chapter 13 Conflicts Among Obligations

Professional judgement becomes most visible when legitimate obligations pull in different directions. Transparency may conflict with confidentiality; timeliness with validation; equal access with approved pre-release coordination; and innovation with reproducibility. The task is not to choose a favorite value but to resolve the conflict under authority and evidence.

Mandatory gates and residual balancing

The first distinction is between mandatory gates and residual considerations. If there is no authority to use the data, expected public benefit does not cure the defect. If a proposed environment cannot protect statistical secrecy, speed does not compensate. Once mandatory gates pass, the decision may balance timeliness, detail, cost, uncertainty and usability.

Non-action also has consequences. Delaying emergency statistics can harm public decisions; refusing a useful accessibility tool can exclude users. A complete options analysis includes the status quo, restricted use, redesign and time-limited provisional use. It does not manufacture balance where a prohibition is clear.

Proportionality

Proportionality asks whether an action is suitable for the legitimate purpose, necessary among available options and balanced in relation to remaining harms. In statistical work, proportionality may justify using less detailed data, a secure environment, a smaller pilot or stronger review rather than abandoning the purpose. The analysis should be explicit about whose burden and risk are being balanced.

AI can generate options and counterarguments, but it may apply generic legal language without jurisdictional basis. The professional uses proportionality as a reasoning structure, cites controlling local authority where it is a legal test and labels broader professional reasoning accurately.

Documenting disagreement

Reasonable professionals may weigh residual uncertainty differently. The institution should record the material disagreement, evidence, decision authority and conditions for review. Suppressing dissent weakens assurance; treating every disagreement as a veto can prevent action. Structured peer challenge and an escalation rule create a governable path.

A decision should be revisited when facts, law, system configuration, performance or intended use change. Review dates and triggers make judgement responsive without making it arbitrary.

Conflict	False solution	Defensible approach
Transparency vs confidentiality	Publish everything or disclose nothing	Explain methods and limits while protecting unit information
Timeliness vs quality	Skip checks to meet deadline	Use approved provisional status, critical checks and revision plan

Conflict	False solution	Defensible approach
Innovation vs accountability	Pilot without ownership	Bounded pilot with owner, benchmark, stopping rule and evidence
Access vs security	Block all users or open all data	Role-based access and safe output pathways

Worked professional situation

During a disease outbreak, Kintaba needs rapid weekly mortality indicators. Complete cause coding is unavailable. The team considers an AI-assisted provisional classifier. Confidentiality and authority gates pass in a secure environment, but quality is uncertain. The decision permits a provisional aggregate with conservative categories, expert review of high-impact cases, visible uncertainty, a revision schedule and a stopping rule if disagreement exceeds tolerance. Timeliness is improved without presenting provisional evidence as final truth.

Verification and decision checks

- Separate mandatory gates from residual balancing.
- Include non-action, restricted use and redesign among the options.
- Identify whose risk and burden are considered.
- Set review triggers when the decision depends on changing facts.

Study questions

43. Which obligations in your setting function as mandatory gates?
44. How can a provisional release be more transparent than a delayed final release?
45. What should a memorandum record when reviewers disagree?

Chapter sources [1], [2], [4], [8]

Chapter 14 Professional Obligations Across the Statistical Lifecycle

Professional duties do not begin at publication. Decisions in planning, source acquisition, collection, processing, analysis and documentation shape which populations are visible, what errors can be detected and who can answer for the final product.

Planning sources and collection

At planning stage, impartiality requires a legitimate information need and attention to affected users. Confidentiality requires purpose and collection authority. Transparency requires clear concepts and respondent information. Quality requires a design capable of supporting the intended use. Accountability requires named owners and review points.

AI may support questionnaire review, stakeholder synthesis or test-case generation. The team should preserve source evidence, check invented requirements and ensure that generated questions do not expand collection beyond authorized need. Collection support must not expose respondent information through unapproved channels.

Processing analysis and quality assurance

Coding, editing, imputation, linkage and anomaly detection can alter records and estimates. Professional control requires benchmarks, reproducible transformations, case-level evidence and rules for uncertainty. AI should not silently convert ambiguity into certainty. When records are changed, the workflow should distinguish suggestion, analyst decision and approved transformation.

Analysis and narrative must maintain conceptual coherence, denominators, revisions and uncertainty. Generated code should be tested; generated interpretation should be compared with tables and methods. Quality assurance should include failure-provoking cases rather than only confirming expected behavior.

Dissemination documentation and learning

Dissemination brings equal access, accessibility, embargo, confidentiality and misuse correction obligations to the foreground. AI can support translation and user service, but responses must preserve approved numbers and meaning. Public chat interfaces require controlled sources, refusal boundaries and monitoring.

Documentation turns one-time judgement into institutional capability. Prompt patterns, decision records, error taxonomies and correction logs should be versioned and reviewed. Retention must remain proportionate and authorized. Learning systems should preserve evidence without accumulating confidential material unnecessarily.

Lifecycle stage	AI-supported task	Professional control
Planning	Need synthesis and risk scenarios	Authority, affected users and concept approval

Lifecycle stage	AI-supported task	Professional control
Acquisition	Metadata extraction and comparison	Purpose, access and source-owner verification
Collection	Guidance and issue triage	Approved channel and escalation
Processing	Coding and anomaly suggestions	Benchmark, provenance and reviewer decision
Analysis	Code and interpretation assistance	Independent tests and uncertainty
Dissemination	Drafting, translation and user response	Approved facts, confidentiality and release authority
Documentation	Methods notes and reusable procedures	Version, retention and review

Worked professional situation

A survey team introduces AI assistance only at report drafting and concludes that governance is simple because no microdata are uploaded. Review shows that the model receives unpublished tables under embargo and drafts comparisons using outdated denominators. The lifecycle analysis identifies confidentiality and quality obligations at dissemination, introduces a controlled source package, locks approved numbers and requires table-to-text consistency tests.

Verification and decision checks

- Map obligations at every lifecycle stage touched by the workflow.
- Preserve the distinction between AI suggestion and authorized statistical transformation.
- Test dissemination outputs against approved numbers and meaning.
- Version reusable procedures and review them after material changes.

Study questions

46. Where in the lifecycle can an apparently low-risk drafting tool become consequential?
47. Which processing decisions require case-level evidence?
48. How should correction logs inform future training and monitoring?

Chapter sources [1], [2], [5], [8]

Chapter 15 A Method for Professional Scenario Analysis

Scenario analysis converts broad principles into a repeatable decision process. It should reveal uncertainty and conflict rather than force a predetermined answer. The method used here has nine elements: decision, authority, facts, information, obligations, affected parties, options, verification and disposition.

Frame the decision precisely

A scenario should begin with the decision requested, not a broad topic. 'Is AI ethical?' is not assessable. 'May the survey division use the approved language model to summarize redacted field reports for weekly issue triage?' identifies a task, environment, data condition, purpose and cadence. The decision should state excluded uses and the authority who must act.

The analyst then identifies facts and unknowns. A fact is supported by evidence in the record. An inference follows from facts but remains an interpretation. An assumption fills missing information and should be tested. A disputed claim requires resolution or explicit treatment in the decision.

Map obligations and consequences

For each obligation, the analyst asks what triggers it, what evidence is relevant and what failure would mean. Affected parties include respondents, data providers, users, statistical staff, decision makers and groups represented or omitted by the product. Consequences include privacy harm, distorted estimates, unequal access, loss of trust, operational burden and missed public value.

Options should be operational. 'Use responsibly' is not an option. An option specifies scope, environment, review, evidence, time limit and stopping rule. At least one restricted or redesigned option should be considered when the purpose is legitimate and a full deployment is unsafe.

Disposition and follow-through

The five standard dispositions are accept, condition, restrict, reject and escalate. Accept means the evidence supports the use under normal controls. Condition requires completion of specified safeguards. Restrict limits data, cases, users or decisions. Reject means the proposed use should not proceed. Escalate means the analyst lacks authority or essential interpretation.

A disposition is incomplete without owners, evidence of completion, review date and incident or correction route. The memorandum should also state what would change the decision. This makes professional judgement revisable without making it unstable.

Element	Core question	Output
Decision	What exact action is requested?	Bounded decision statement
Authority	Who may decide and under what mandate?	Authority map
Facts	What is verified, inferred, assumed or disputed?	Claim ledger
Information	What may enter which environment?	Handling decision

Element	Core question	Output
Obligations	Which duties apply and conflict?	Obligation map
Options	What feasible actions and non-actions exist?	Option set
Verification	What evidence challenges the preferred answer?	Test and correction log
Disposition	Accept, condition, restrict, reject or escalate?	Judgement and safeguards

Worked professional situation

The price-statistics unit proposes using AI to draft explanations of unusual movements from analyst notes. Scenario analysis identifies that notes may contain confidential business context, the release is market sensitive and causal statements require evidence. The unrestricted option fails. A redesigned option uses a controlled environment, excludes company identifiers, restricts the model to approved tables and public metadata, requires source-linked claims and retains final analyst approval. The decision is conditionally accepted for a monitored pilot.

Verification and decision checks

- Write the exact decision and excluded uses.
- Label facts, inferences, assumptions and disputes separately.
- Consider affected parties and non-action.
- Convert conditions into owners, deadlines and observable completion evidence.

Study questions

49. How does a bounded decision improve AI prompting?
50. What is the difference between restrict and condition?
51. When is escalation the only professionally valid disposition?

Chapter sources [4], [8], [11]

Chapter 16 The Evidence and Obligation Ledger

A ledger prevents fluent narrative from hiding weak evidence. It records material claims, source authority, confidence, contradictions, triggered obligations and the decision consequence. It is both a learning tool and a practical control for AI-assisted analysis.

Claim-level traceability

Each material claim should identify its source, source type, version or date and whether the source directly supports the claim. Quotations require exact verification. Summaries should preserve conditions and exceptions. Where the evidence is incomplete, the ledger records the gap rather than allowing the narrative to imply certainty.

AI-generated source lists are not evidence. Retrieval can accelerate location, but the reviewer must open the source and confirm support. Where sources are dynamic web pages, institutions may retain an authorized snapshot or citation record subject to copyright and records rules.

Obligation mapping

A claim may trigger several duties. The statement that a vendor retains prompts triggers confidentiality, procurement and accountability questions. The ledger links the evidence to obligations, affected parties and required action. This avoids treating principles as detached headings added after the technical analysis.

Conflicts should be explicit. A transparency entry might support fuller method disclosure, while confidentiality evidence limits disclosure of small-area inputs. The decision column records how the conflict is resolved and which authority supports that resolution.

Confidence and contradiction

Confidence is not an AI probability. It is the reviewer's assessment of evidentiary support for the professional decision. A high-quality ledger distinguishes strong authoritative evidence, credible operational evidence, expert judgement and unverified assertion. Contradictions are not averaged away; they are resolved, bounded or escalated.

The ledger should remain usable. Recording every trivial statement creates noise and can increase confidentiality exposure. Materiality is defined by whether the claim could change the decision, safeguard, public interpretation or accountability.

Field	Meaning	Example
Claim	Material proposition affecting decision	Provider retains prompts for 30 days
Source and status	Authority, version and jurisdiction	Tenant contract, clause 8, current
Support	Direct, indirect, contradicted or absent	Direct
Obligation	Duty triggered by the claim	Confidentiality and accountability
Decision effect	How the evidence changes action	Exclude protected data and seek shorter retention

Field	Meaning	Example
Reviewer state	Verified, pending or escalated	Verified by procurement and security

Worked professional situation

An AI assistant asserts that the national statistics law permits data sharing with all government agencies. The ledger records the claim as material, the AI output as non-authoritative and support as absent. Review of the statute shows access is limited to specified statistical purposes and confidentiality conditions. The claim is corrected, the proposed transfer is restricted and legal review is requested for the remaining ambiguity. The ledger prevents the fabricated generalization from entering the memorandum.

Verification and decision checks

- Record only claims material to decision, safeguard or interpretation.
- Verify sources directly and preserve their conditions.
- Link evidence to obligations and decision consequences.
- Resolve, bound or escalate contradictions rather than averaging them.

Study questions

52. How should materiality be defined for an obligation ledger?
53. What source categories should your institution recognize?
54. Why should AI output never be recorded as the authority for a legal claim?

Chapter sources [1], [2], [4], [11]

Chapter 17 Designing the AI Task Specification

A professional prompt is a controlled task specification. It states purpose, role, context, permitted information, definitions, constraints, output schema, evidence requirements, uncertainty treatment and refusal conditions. It makes the desired reasoning inspectable without transferring professional authority to the model.

Context without oversharing

The model needs enough context to distinguish statistical from generic organizational issues, but every additional detail should be justified. Synthetic identifiers, generalized roles and redacted facts can often support scenario analysis. The instruction should explicitly prohibit inference of missing confidential facts and should mark unresolved information for human follow-up.

Definitions matter. Terms such as impartiality, quality and accountability should be tied to the approved framework. Otherwise the model may use broad popular meanings that omit official-statistics requirements. The source pack should identify controlling references and tell the model not to invent authorities.

Output designed for verification

A structured output improves review. Useful fields include decision requested, facts, assumptions, obligations, affected parties, options, missing evidence, risks, preliminary disposition and questions for the accountable reviewer. Every authority claim should include a source identifier or be labelled unverified.

The instruction should request counterarguments and conditions that would reverse the preliminary result. This reduces one-sided completion but does not eliminate bias. The analyst should test alternative framing and compare outputs.

Refusal and escalation

The model should be instructed to refuse legal conclusions, final authorization and use of prohibited data. It should identify when essential authority is missing. Refusal is useful only if the human workflow routes the case appropriately; otherwise users may rephrase until they obtain a desired answer.

Versioning records the instruction, source pack, tool and date. Reuse conditions state which scenarios the task specification covers and when it must be reviewed. A prompt that worked once is not automatically a standard operating procedure.

Specification field	Required content
Purpose	Analyze a bounded scenario and prepare evidence for human judgement
Permitted inputs	Synthetic or approved facts and controlled source identifiers
Prohibited actions	No legal ruling, final approval, invented authority or inference of protected facts

Specification field	Required content
Output	Structured facts, obligations, conflicts, options, gaps and draft memorandum
Evidence	Source identifier for every material normative claim
Challenge	Contrary interpretation, omitted stakeholder and failure condition
Escalation	Mark missing authority or unresolved conflict for named reviewer

Worked professional situation

A weak prompt asks, 'Is it ethical to use AI on survey data?' The improved specification defines a synthetic field-report triage task, the approved environment, excluded identifiers, applicable institutional policy, output fields and a prohibition on final authorization. It asks for missing facts and contrary arguments. The result is longer but easier to verify because the model cannot hide assumptions inside a simple yes or no.

Verification and decision checks

- Minimize context and prohibit inference of missing protected facts.
- Supply approved definitions and controlled source identifiers.
- Require a structure that exposes assumptions and evidence gaps.
- Version the task specification and define reuse boundaries.

Study questions

55. Which prompt fields make professional judgement easier to audit?
56. How can refusal conditions be tested?
57. When does a prompt become a controlled institutional procedure?

Chapter sources [8], [11], [14]

Chapter 18 AI Assisted Analysis and Adversarial Challenge

AI-assisted analysis should be treated as a candidate reasoning product. Verification asks whether its claims are supported; adversarial challenge asks how it could fail. Both are necessary because a response can be internally coherent and still omit decisive evidence or reproduce the user's framing.

Independent verification

The reviewer verifies authorities, facts, calculations, classifications and links using sources independent of the generated answer. Copying the model's citations into a browser is not sufficient if the cited source does not contain the claim. The verification log records what was checked, by whom, what changed and what remains uncertain.

For repeated scenario analysis, test cases can contain known obligation conflicts, fabricated policies, ambiguous authority and confidentiality traps. The objective is not deterministic wording but consistent detection of material issues and appropriate escalation.

Adversarial prompts

Useful challenges ask what affected party is missing, which fact would reverse the recommendation, whether the conclusion changes under a stricter confidentiality assumption, what authority has been inferred and what plausible harm has been discounted. A separate review prompt may critique the draft, but it should not be treated as independent evidence when it uses the same model and sources.

Prompt injection and malicious content also matter when the system retrieves external documents. A document may contain instructions to reveal data or ignore policy. The system and reviewer should treat retrieved content as evidence, not as executable authority.

Variability and stability

Generative outputs vary across runs and model versions. Material tasks should be repeated or tested with controlled variations to identify unstable conclusions. Stability is not correctness, but instability can reveal sensitivity to framing. Where a disposition changes across minor prompt variations, the workflow should rely more heavily on structured rules and human analysis.

Corrections should improve the procedure, not only the current text. Error taxonomies, revised prompts and regression cases help prevent recurrence. Tool changes trigger targeted retesting of material scenarios.

Challenge	Failure sought	Reviewer action
Opposite conclusion	Confirmation and framing bias	Compare evidence required for each conclusion
Missing stakeholder	Narrow affected-user analysis	Add burden, access and consequence evidence

Challenge	Failure sought	Reviewer action
Stricter data assumption	Confidentiality underclassification	Re-evaluate environment and minimization
Authority audit	Invented or misapplied rule	Open source and escalate interpretation
Prompt variation	Unstable disposition	Restrict use and strengthen structured controls

Worked professional situation

A model recommends accepting a vendor because its privacy page says customer data are not used for training. An adversarial challenge asks about logs, support access, subprocessors, retrieval storage, deletion and audit. The model identifies gaps but invents a standard audit clause. The reviewer verifies the contract, removes the invented clause and changes the disposition from accept to condition pending procurement and security evidence.

Verification and decision checks

- Verify material claims outside the generated response.
- Use challenges that could change the decision, not cosmetic critique.
- Treat same-model critique as assistance rather than independent evidence.
- Convert recurring failures into regression tests and workflow changes.

Study questions

58. Why is a second model response not independent verification?
59. Which prompt variations should be included in a stability test?
60. How should retrieved malicious instructions be handled?

Chapter sources [8], [11], [14]

Chapter 19 The Professional Judgement Memorandum

The professional judgement memorandum is the module's required product. It provides a concise, reviewable bridge between evidence and institutional action. Its purpose is not to display every thought but to make the material reasoning, authority, uncertainty and accountability visible.

Structure and audience

The memorandum opens with the exact decision requested and the authority to decide. It then states verified facts and disputed points, applicable obligations, information-handling conditions, options, verification, recommendation, safeguards and review. The intended decision body determines length and language, but compression must not erase material uncertainty.

Facts, interpretations and recommendations should be distinguishable. Sources may be cited through a compact ledger or annex. The memorandum should not reproduce confidential scenario details beyond the recipient's authorization.

Decision language

The disposition uses explicit verbs: accept, condition, restrict, reject or escalate. Conditions state owner, due date and completion evidence. Restrictions define data, users, cases, outputs or time period. Rejection states which gate failed and what redesign might permit reconsideration. Escalation identifies the unresolved question and authorized recipient.

Residual risk should be stated, not buried in limitations. The decision body needs to know what could still fail, how it will be detected and who can stop operation. A monitoring promise without indicator or owner is not a safeguard.

AI assistance and authorship

AI may help organize the ledger, compare options and draft prose. The memorandum remains the professional product of the named author and approver. The record should preserve the material task specification, original output where permitted, checks and corrections. Public disclosure of AI assistance is determined by materiality and policy.

Quality-adjusted time is reported when efficiency is claimed. The comparison includes generation, verification and rework and confirms that both routes meet the same threshold. The institution should avoid incentives that reward faster memoranda while ignoring weaker evidence.

Section	Required answer	Common defect
Decision	What exact action is requested and by whom?	Vague topic rather than decision
Facts	What is verified and what remains uncertain?	Generated assertions treated as facts
Obligations	Which duties and authorities apply?	Generic ethics language
Options	What feasible actions and consequences exist?	Approve or reject only

Section	Required answer	Common defect
Judgement	What disposition and why?	Recommendation detached from gates
Safeguards	Who does what, by when and how verified?	Unowned monitoring promise
Review	What trigger or date reopens the decision?	Permanent approval despite changing system

Worked professional situation

A draft memorandum states that an AI classification tool is 'generally ethical if monitored.' Revision converts this into a restricted pilot decision: public synthetic and approved de-identified inputs only; no automatic final codes; 10 percent stratified review plus full review of low-confidence and minority-language cases; monthly subgroup analysis; immediate suspension on confidentiality incident or material tolerance breach; methodologist owner; review after two production cycles. The revised memorandum is actionable and auditable.

Verification and decision checks

- Put the decision and authority first.
- Separate facts, interpretations, assumptions and unresolved issues.
- Write conditions as owned, testable actions.
- State residual risk, stopping rules and review triggers.

Study questions

61. What information belongs in an annex rather than the decision body?
62. How can a memorandum remain concise without hiding uncertainty?
63. Which AI interaction evidence should be retained and for how long?

Chapter sources [4], [8], [11]

Chapter 20 Worked Case Publication Pressure

This case integrates impartiality, professional independence, quality, transparency and accountability. It demonstrates why a statistically correct table can still be surrounded by an unacceptable release process.

Scenario

Kintaba's labour force survey release is scheduled for Friday at 10:00. The unemployment rate rises from 8.1 to 8.4 percent, but the change is not statistically significant. On Wednesday, a ministerial adviser asks the NSO to delay release until Monday and requests an AI-generated note emphasizing youth employment gains. The published calendar allows changes only for documented technical failure or national emergency. The adviser has pre-release access under a restricted briefing arrangement but no release authority.

The drafting assistant receives approved tables and public metadata. Its first output describes a 'sharp deterioration' in unemployment and recommends delay to ensure responsible communication. Neither conclusion is supported by the method or policy.

Analysis

Authority is clear: the designated statistical authority controls release timing. The adviser's request is evidence of potential interference but does not itself prove improper intent. The facts distinguish the numerical increase from its uncertainty. Transparency requires explanation of the estimate and sampling variability; impartiality requires balanced treatment of youth gains and unemployment uncertainty; accountability requires a documented release decision.

The AI output fails because it converts a non-significant change into a strong trend and invents a policy rationale. The analyst verifies the release calendar, estimate, standard error and approved language. Confidentiality is not the primary issue because inputs are approved, although embargo controls still apply.

Disposition and memorandum

The defensible disposition is reject the delay request and proceed under the release calendar, while accepting the legitimate need for clear explanation. The release note states that the point estimate increased but evidence is insufficient to conclude a quarter-to-quarter change; it reports the youth indicator using the same approved comparisons. The request, policy check, AI draft and corrections are retained in the controlled record.

A separate institutional action reviews whether the pre-release briefing process and AI drafting instruction create pressure or ambiguity. This distinguishes correction of the current release from prevention of recurrence.

Memorandum field	Case entry
Decision	Whether to delay the scheduled release and adopt requested framing

Memorandum field	Case entry
Authority	Chief statistician or delegated release authority under published calendar
Failed AI claims	Sharp deterioration; delay justified by responsible communication
Obligations	Impartiality, independence, quality, transparency, equal access and accountability
Disposition	Reject delay; correct narrative; release on schedule
Safeguards	Approved significance language, source comparison, controlled record and process review

Worked professional situation

The baseline manual analysis takes 38 minutes. The AI-assisted draft takes 4 minutes, verification 18 minutes and correction 9 minutes, totaling 31 minutes. Both final products meet the threshold, so the net quality-adjusted saving is 7 minutes, not 34. More importantly, the error log shows that the original prompt needs stronger significance and policy constraints. The reusable value lies in the corrected procedure, not the first fluent draft.

Verification and decision checks

- Verify release authority and exceptions directly.
- Distinguish point-estimate movement from statistical evidence of change.
- Preserve balanced comparisons and approved uncertainty language.
- Record the interference concern through the appropriate institutional route.

Study questions

64. Would the decision change if a verified technical failure occurred?
65. How should the analyst describe the adviser's request without inferring motive?
66. What regression tests should be added to the drafting assistant?

Chapter sources [1], [3], [5], [8]

Chapter 21 Worked Case Confidential Administrative Data

This case integrates confidentiality, purpose limitation, quality, supplier evidence, transparency and accountability. It shows why removing direct identifiers does not by itself authorize AI use.

Scenario

Kintaba's education-statistics unit receives administrative school records under a statistical agreement. Analysts want an AI service to classify free-text reasons for dropout. Names and student numbers are removed, but text may contain village, disability, household event and staff information. The vendor states that enterprise data are not used for model training but retains prompts for thirty days for abuse monitoring. The agreement permits statistical production and quality improvement but does not mention external processing.

A pilot on 500 records achieves 87 percent agreement with human coding. Performance for disability-related descriptions is uncertain because only 12 benchmark cases exist.

Analysis

The data remain potentially identifiable through rare combinations and free text. Removing names is insufficient. Authority for external processing is unresolved, and retention creates an additional disclosure pathway. The average agreement does not establish fitness for disability-related coding. Transparency to data providers and users may also require review, but it cannot substitute for access authority.

Options include rejection of external use, a secure local model, stronger transformation, synthetic prompt development or an amended approved environment. The AI system can help inventory sensitive patterns on synthetic examples but should not receive the actual records until authority and environment pass.

Disposition and memorandum

The defensible disposition is escalate the authority question and restrict current work to synthetic development. If authority is established, a controlled pilot should minimize or transform text, disable unnecessary retention where possible, apply access logging, expand the benchmark and route disability-related and low-confidence cases to expert review. The final decision belongs to the designated statistical, confidentiality, legal and security authorities.

The memorandum states that the statistical purpose is legitimate but the proposed technical route is not yet authorized. This distinction prevents a useful objective from being confused with approval of a specific tool.

Gate	Status	Reason
Statistical authority	Conditional	Purpose is statistical, but external processing is unresolved
Confidentiality	Fail pending redesign	Free text and retention create disclosure risk

Gate	Status	Reason
Quality	Partial	Average agreement passes; critical subgroup evidence is weak
Oversight	Design required	Expert review and stopping rules not yet specified
Security continuity	Unknown	Architecture, access and deletion evidence incomplete

Worked professional situation

The initial manual coding takes 210 minutes. The AI run takes 12 minutes, but a valid comparison cannot yet be claimed because confidentiality and subgroup quality gates fail. Even if review took only 50 minutes, the time saving has no professional value until the method is authorized and meets the same quality standard. This is the central discipline of quality-adjusted efficiency.

Verification and decision checks

- Treat free text and rare combinations as potentially identifiable.
- Separate legitimate purpose from authorization of the proposed environment.
- Do not accept average performance when a consequential subgroup is unresolved.
- Give time savings no credit when mandatory gates fail.

Study questions

67. What transformations could reduce disclosure while preserving coding value?
68. How should the team expand a rare-category benchmark?
69. Which authorities must approve external processing in your jurisdiction?

Chapter sources [1], [2], [5], [9], [14]

Chapter 22 Synthesis Assessment and Workplace Transfer

Professional obligation is the disciplined capacity to decide under authority, evidence and uncertainty. The competent official statistician neither accepts AI output because it is fluent nor rejects AI because it is unfamiliar. They define the task, protect information, test evidence, record judgement and make reuse conditional on demonstrated control.

Integrated competence

The five obligations reinforce one another. Impartiality protects purpose and treatment of evidence. Confidentiality protects information and the relationships through which it is obtained. Transparency supports interpretation and challenge. Quality connects methods to intended use. Accountability assigns decisions, safeguards and correction. A defect in one can weaken the entire product.

The module's six learning outcomes are demonstrated through a complete scenario record and memorandum. The learner must identify authority, trace evidence, classify information, challenge AI reasoning, issue a disposition and defend it. Knowledge of principle names without performance is insufficient.

Workplace transfer

The reusable artifact is not a generic prompt. It is a controlled package: scenario intake, authority register, information screen, obligation ledger, task specification, verification checklist, memorandum template and review schedule. Each institution localizes law, roles, approved tools and incident routes without weakening the canonical reasoning.

Transfer should begin with low-consequence synthetic or public cases, then supervised approved work. The institution monitors error, unsafe tendencies, review burden, time and recurring ambiguity. A successful pilot results in a controlled procedure; an unsuccessful pilot results in documented learning and a restrict, redesign or stop decision.

Connection to the pathway

OS-A01 located the actors, sources, users, flows and authority of the national information ecosystem. OS-A02 established a task-based understanding of AI mechanisms and failure modes. OS-A03 adds professional judgement. OS-A04 will use these foundations to frame bounded use cases with explicit users, benefits, consequences and boundaries.

Professional obligations recur throughout the pathway. Block C deepens quality and source fitness. Block D develops authority, confidentiality, suppliers, oversight and incidents. Block E develops statistical assurance. OS-A03 provides the common decision language that keeps those specialist controls connected to the responsibility of official statisticians.

Disposition	Use when	Required record
Accept	All gates pass and residual risk is within authority	Approval, normal controls and review

Disposition	Use when	Required record
Condition	Use may proceed after specified safeguards	Owner, deadline and completion evidence
Restrict	Only bounded data, users, cases or decisions are supported	Scope and stopping rule
Reject	Mandatory gate fails or redesign is not proportionate	Failure evidence and reconsideration condition
Escalate	Authority, legal meaning or risk acceptance exceeds role	Question, evidence and authorized recipient

Worked professional situation

For the final assessment, the learner receives a new Kintaba scenario containing incomplete authority, mixed public and protected evidence, an AI-generated preliminary memorandum and a deadline. The learner must decide what may enter the tool, correct false claims, construct the ledger, test alternatives, issue the memorandum and defend the disposition. Any unauthorized disclosure, fabricated authority, concealed material error, bypassed gate or delegation of the final decision to AI results in not yet competent regardless of the numerical score.

Verification and decision checks

- Use the same professional acceptance standard for baseline and AI-assisted work.
- Retain evidence sufficient for review without retaining unnecessary protected information.
- Convert successful practice into a versioned controlled procedure.
- Reopen decisions when authority, data, system, performance or intended use changes.

Study questions

70. Which component of the workplace package requires the most localization?
71. How will you measure whether AI improves judgement rather than only drafting speed?
72. What evidence would justify moving from supervised practice to routine use?

Chapter sources [1], [2], [4], [8], [11]

Professional Judgement Memorandum Template

Use this structure for the module practical and adapt it only where institutional policy requires. Keep the main memorandum concise and place detailed evidence in controlled annexes.

Section	Required content
Decision requested	Action, product, users, deadline, decision owner and excluded decisions
Verified facts	Material facts, source identifiers, confidence, disputed points and missing evidence
Authority	Mandate, policy, delegation, reviewers and matters requiring escalation
Information handling	Classification, permitted environment, minimization, retention and exclusions
Professional obligations	Impartiality, confidentiality, transparency, quality and accountability analysis
Options	Accept, condition, restrict, reject, escalate and non-action consequences
Verification	Claims, authorities, calculations, outputs, challenge tests and corrections
Judgement	Disposition and concise evidence-based rationale
Safeguards	Owner, due date, completion evidence, monitoring and stopping rule
Review	Residual uncertainty, review date, change triggers and correction route
Efficiency	Baseline and AI-assisted time, verification, rework and threshold result

Consolidated Assessment Rubric

Criterion	Weight	Competent evidence	Critical failure
Authority and boundary	15	Decision, mandate, users and excluded actions are explicit.	Acts beyond authority or bypasses a mandatory gate.
Evidence and provenance	15	Facts and obligations are traceable; uncertainty is stated.	Fabricates or conceals material evidence.
Impartiality and independence	15	Influence, conflicts and equal-treatment issues are addressed.	Accepts improper influence or discriminatory access.

Criterion	Weight	Competent evidence	Critical failure
Confidentiality and safe AI use	20	Classification, minimization, environment and retention are correct.	Uses or discloses protected information without authorization.
Quality and verification	15	Checks and limitations fit the intended use.	Approves a failed result or claims checks not performed.
Professional judgement	15	Disposition is proportionate, reasoned and escalated where needed.	Delegates the consequential decision to AI.
Communication and record	5	Memorandum is clear, reviewable and decision-ready.	Materially obscures limitation or responsibility.

Competence requires at least 70 out of 100, no critical failure, and a competent rating in confidentiality, verification and professional judgement. Reassessment requires corrected work and an oral explanation of the original failure mechanism.

Glossary

Term	Definition
Accountability	The assignment and exercise of authority, review, answerability, correction and consequences.
AI-assisted workflow	A bounded process in which AI supports tasks while authorized humans retain professional decisions.
Confidentiality	Protection of information obtained for statistical purposes against unauthorized access, use or disclosure.
Disposition	The recorded decision to accept, condition, restrict, reject or escalate.
Impartiality	Objective production and dissemination based on professional and statistical considerations.
Professional judgement	A reasoned decision made under authority using evidence, principles, uncertainty and consequence.
Quality-adjusted time	Workflow time compared only after outputs satisfy the same approved quality threshold.
Residual risk	Risk remaining after approved safeguards and accepted by the authorized decision level.
Transparency	Provision of information needed to understand, interpret, review and correct a statistical process or product.

References

- [1] United Nations General Assembly. Fundamental Principles of Official Statistics. Resolution 68/261, 2014, with official UN Statistics Division implementation resources.
- [2] United Nations Statistics Division. United Nations National Quality Assurance Frameworks Manual for Official Statistics. 2019.
- [3] European Statistical System Committee. European Statistics Code of Practice. Revised edition 2017, including the Quality Assurance Framework.
- [4] International Statistical Institute. Declaration on Professional Ethics. Current official edition and supporting documentation.
- [5] United Nations Statistics Division. Handbook on Management and Organization of National Statistical Systems, especially the basis and organization of official statistics chapters.
- [6] Organisation for Economic Co-operation and Development. Recommendation of the Council on Good Statistical Practice. OECD/LEGAL/0417.
- [7] Principles Governing International Statistical Activities. United Nations Committee for the Coordination of Statistical Activities.
- [8] SRAI. Phase 2A Curriculum Blueprint and Competency Architecture. Version 1.0, September 2026.
- [9] SRAI. Phase 2B Module and Laboratory Production Standard. Version 1.0, September 2026.
- [10] SRAI. Official Statistics and AI Pathway Production Reference. Version 1.3, 20 September 2026.
- [11] National and institutional statistical law, confidentiality, release, information-security, records, procurement and incident policies applicable to the learner's jurisdiction.
- [12] UNECE. Guidance and work on machine learning and artificial intelligence in official statistics, including quality and responsible-use considerations.
- [13] International Statistical Institute and International Association for Official Statistics. Statements concerning professional independence and integrity of official statistics.
- [14] Applicable approved tool documentation, data-processing terms, security architecture and institutional clearance record for the actual AI configuration.

Source Status Note

References [1] through [7], [12] and [13] provide international normative or professional guidance. References [8] through [10] control this pathway's curriculum and production. Reference [11] controls localized authority within its jurisdiction. Reference [14] is configuration-specific evidence. Users must verify current versions and local applicability before making consequential decisions.